

# Training PINNs

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ICML 2024 • PMLR 235 • Contact: pratikr@stanford.edu

Wave PDE training curves: Adam vs Adam+L-BFGS vs NNCG

Adam converges slowly; Adam+L-BFGS stalls ~40k steps; NNCG after Adam+L-BFGS further improves.

## ABSTRACT

- PINN loss is ill-conditioned due to differential operators.
- Adam+L-BFGS beats Adam or L-BFGS; NNCG further improves.
- Theory links operator and loss conditioning; supports hybrids.

OPTIMIZER

Adam+L-BFGS best

NEW METHOD

NNCG

EFFECT

Lower loss & L2RE

## INTRODUCTION

- PINNs solve PDEs via residual, boundary, and initial losses.
- Mesh-free promise vs hard optimization from ill-conditioning.
- We study the loss landscape and improve training with hybrids.

## KEY FINDING

- Single sentence: Even after Adam+L-BFGS stalls, a brief NNCG phase slashes loss and L2RE by another order of magnitude.
- Explanation: L-BFGS halts under strong Wolfe before a critical point ( $\text{grad} \approx 1\text{e-}2\text{--}1\text{e-}3$ ); NNCG's damped Newton steps with Armijo and Hessian-vector products exploit remaining curvature to make decisive progress.

## CONTRIBUTIONS

- Hessian spectra reveal severe ill-conditioning; quasi-Newton improves  $\geq 1000\times$ .
- Adam+L-BFGS outperforms Adam or L-BFGS across problems.
- Accurate solutions demand near-zero loss; NNCG boosts further.
- Theory: operator ill-conditioning  $\Rightarrow$  loss ill-conditioning; hybrids justified.

## SETUP

- Model  $u(x; w)$  with tanh MLP; minimize residual and BC/IC losses.
- Interpolation (loss $\approx 0$ ) is feasible and desirable for generalization.
- Common pipeline: Adam then L-BFGS.

 Training pipeline: sampling  $\rightarrow$  model  $\rightarrow$  losses  $\rightarrow$  optimizer switches

Pipeline schematic: dataset sampling  $\rightarrow$  PINN (MLP)  $\rightarrow$  residual/BC/IC losses  $\rightarrow$  Adam  $\rightarrow$  L-BFGS ( $\rightarrow$  NNCG).

## METHODS

- Convection, wave PDEs; reaction ODE (challenging settings).
- MLPs: 3 layers; widths 50/100/200/400; Xavier init.
- Sampling: 10k residual; dense IC/BC; metric: L2RE.
- Optimizers: tuned Adam; L-BFGS (lr=1.0, m=100); switches at 1k/11k/31k; total 41k iters.
- Env: PyTorch 2.0, Python 3.10, Titan V, CUDA 11.8.

## LOSS VS ERROR

- Lower training loss  $\rightarrow$  lower L2RE across PDEs.
- Convection: loss  $1e-3 \rightarrow$  L2RE  $\approx 1e-1$ ;  $1e-5 \rightarrow \approx 1e-2$ .

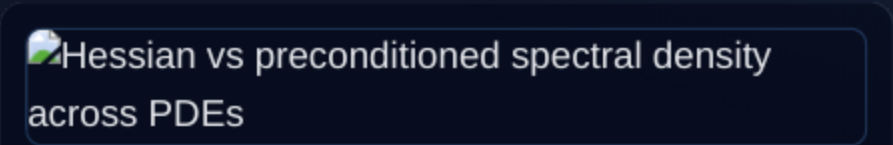
 Scatter: final L2RE vs final loss across runs

Lower loss correlates with better accuracy (Figure 2).



## ILL-CONDITIONING

- Hessian spectra: large outliers and mass near 0.
- Residual term (with D) dominates ill-conditioning.
- L-BFGS preconditioning cuts top eigenvalues  $\geq 1e3$ .



Hessian vs preconditioned spectral density across PDEs

Convection/Reaction/Wave: severe ill-conditioning; quasi-Newton improves (Figure 3 top).



Component-wise spectra: residual, IC, BC

Residual dominates; all components improved by L-BFGS (Figure 3 bottom).

## OPTIMIZERS

- Adam+L-BFGS achieves lowest loss and L2RE across widths and PDEs.
- Examples (Table 1): Convection L2RE  $5.96e-2$  (Adam)  $\rightarrow$   $4.19e-3$  (Adam+L-BFGS); Wave  $3.35e-1$  (L-BFGS)  $\rightarrow$   $5.52e-2$ .

ACROSS PDES

Combined wins

REASON

Avoid saddles + faster local rate

EXCEPTION

Reaction milder

## HYBRID + NNCG

- First-order avoids saddles but slows when  $\kappa$  is large; quasi-/second-order speed locally.
- Adam  $\rightarrow$  L-BFGS: escape saddles, then reduce  $\kappa$ ; if stalled, switch to NNCG.
- NNCG: damped Newton with Armijo; Hessian-free via Hvps; Nyström-PCG preconditioning.



Diagram: Adam  $\rightarrow$  L-BFGS  $\rightarrow$  NNCG with line searches and preconditioning

Pipeline with NNCG fine-tuning for high precision.

## UNDER-OPTIMIZED

- L-BFGS with strong Wolfe may pick step size 0 and stop before a critical point.

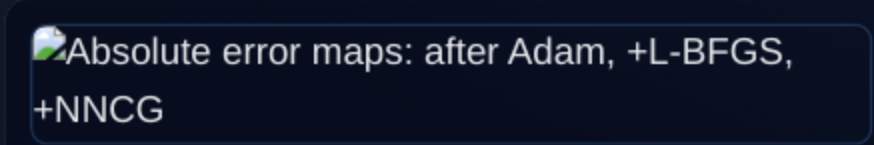


Training curves showing L-BFGS early stop and nonzero gradients

Runs stall with grad norms  $\approx 1e-2$ – $1e-3$  (Figure 4/9).

## NNCG RESULTS

- After Adam+L-BFGS, NNCG reduces loss  $>10\times$  and lowers gradients; GD gives no gains.



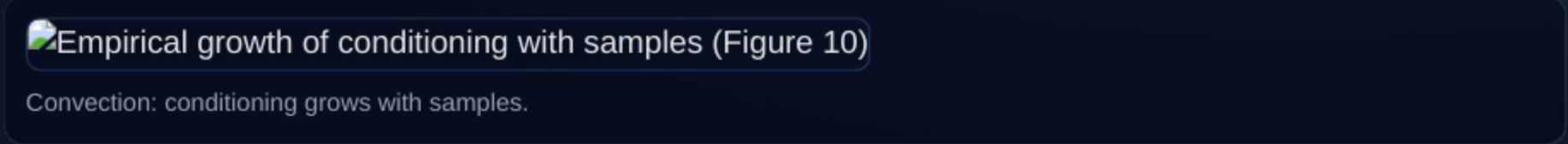
Progressive error reduction (Figure 5).



NNCG best; GD  $\approx$  no change (Table 2).

## OPERATOR $\rightarrow$ LOSS

- Ill-conditioned D induces ill-conditioned PINN loss.
- Theorem 8.4 (informal): spectrum of  $A\cdot K_\infty(w_\star)$  controls conditioning; polynomial decay  $\Rightarrow \kappa$  grows with  $n_{\text{res}}$ .



Convection: conditioning grows with samples.

## GDND RATE

- GDND: switch from GD to damped Newton; Phase II has linear rate independent of  $\kappa$  (Theorem 8.5).



Local PL  $\star$  enables rapid Phase II convergence.